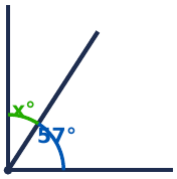
Unit 4 Homework — Angle Relationships — Answer Key

Complete independently · Complementary/Supplementary · Vertical · Parallel Lines · Triangles

A. Complementary, Supplementary & Linear Pair

Problem 1 Score 3

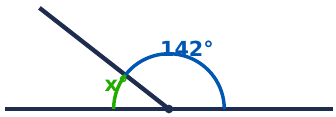
The two angles shown are complementary. Find x.



Answer: 33

Problem 2 Score 3

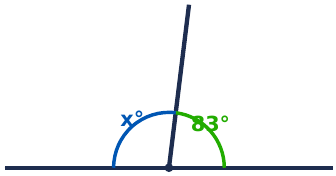
The two angles shown are supplementary. Find x.



Answer: 38

Problem 3 Score 3

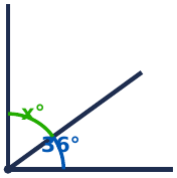
These two angles form a linear pair. Find x.



Answer: 97

Problem 4 Score 4

The two angles shown are complementary. Find x.

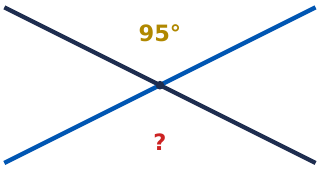


Answer: 54

B. Vertical Angles

Problem 1 Score 3

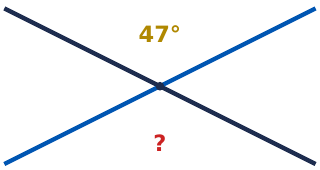
The lines shown cross at a point. Find x (vertical angles).



Answer: 95

Problem 2 Score 4

The lines shown cross at a point. Find x (vertical angles).

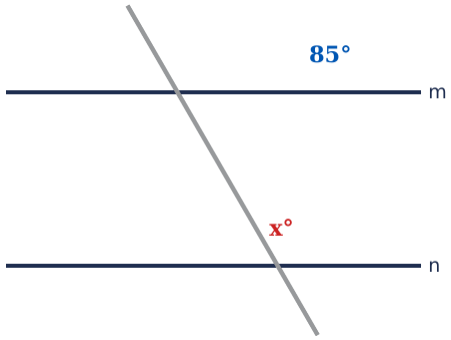


Answer: 47

C. Parallel Lines & Transversals

Problem 1 Score 3

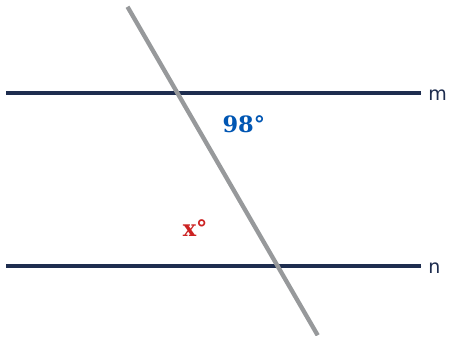
Lines m and n are parallel. The marked angles are corresponding angles. Find x.



Answer: 85

Problem 2 Score 4

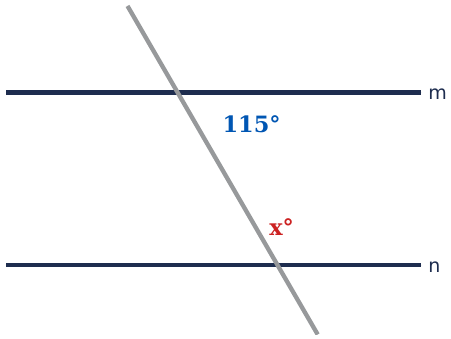
Lines m and n are parallel. The marked angles are alternate interior angles. Find x.



Answer: 98

Problem 3 Score 4

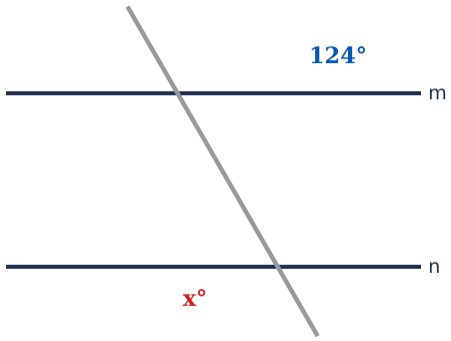
Lines m and n are parallel. The marked angles are same-side interior angles. Find x.



Answer: 65

Problem 4 Score 4

Lines m and n are parallel. The marked angles are alternate exterior angles. Find x.



Answer: 124

Problem 5 Score 5

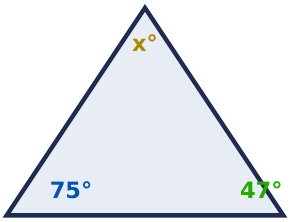
Two parallel lines are cut by a transversal. Two same-side interior angles are (4x + 10)° and (x + 20)°. Find the LARGER angle, in degrees.

Answer: 130

D. Triangle Angle Sum & Exterior Angle Theorem

Problem 1 Score 3

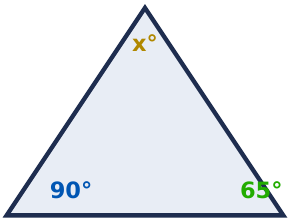
Find x in the triangle shown.



Answer: 58

Problem 2 Score 4

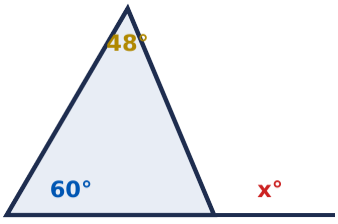
Find x in the triangle shown.



Answer: 25

Problem 3 Score 4

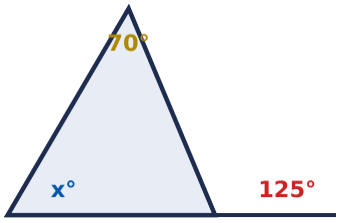
The exterior angle theorem: find x (the exterior angle).



Answer: 108

Problem 4 Score 5

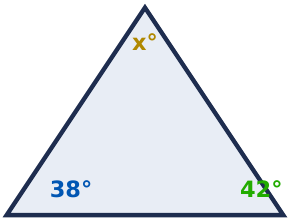
The exterior angle is 125°. One remote interior angle is 70°. Find x.



Answer: 55

Problem 5 Score 5

Find x in the triangle shown.



Answer: 100

Problem 6 Score 5

Two angles of a triangle are (3x)° and (x + 20)°. The exterior angle at the third vertex is 120°. Find the larger of those two angles, in degrees.

Answer: 75

Problem 7 Score 5

Why does an exterior angle of a triangle equal the two remote interior angles added together?

A. Every exterior angle is 120°.

B. The exterior angle and the angle next to it add to 180°. The three inside angles also add to 180°. So the exterior angle equals the other two.

C. The exterior angle is always the biggest angle.

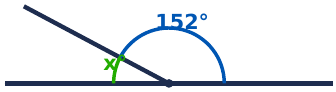
D. The exterior angle and the remote angles are vertical angles.

Answer: B

E. Mixed Review

Problem 1 Score 4

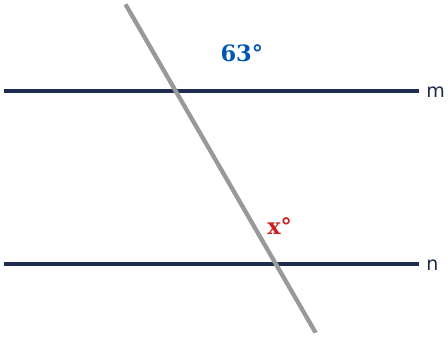
The two angles shown are supplementary. Find x.



Answer: 28

Problem 2 Score 4

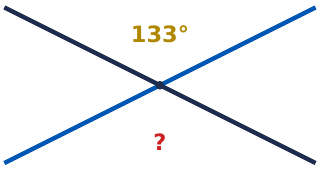
Lines m and n are parallel. The marked angles are corresponding angles. Find x.



Answer: 63

Problem 3 Score 4

The lines shown cross at a point. Find x (vertical angles).



Answer: 133